

VitalX – Decentralized Health Record Management System

A Blockchain-Based Healthcare Project by SPPU

Introduction: Revolutionizing Healthcare Data



Decentralized Management

VitalX is a decentralized health record management system, eliminating central points of control.



Secure Data Storage

Leveraging Blockchain and IPFS, medical data is stored with unparalleled security and integrity.



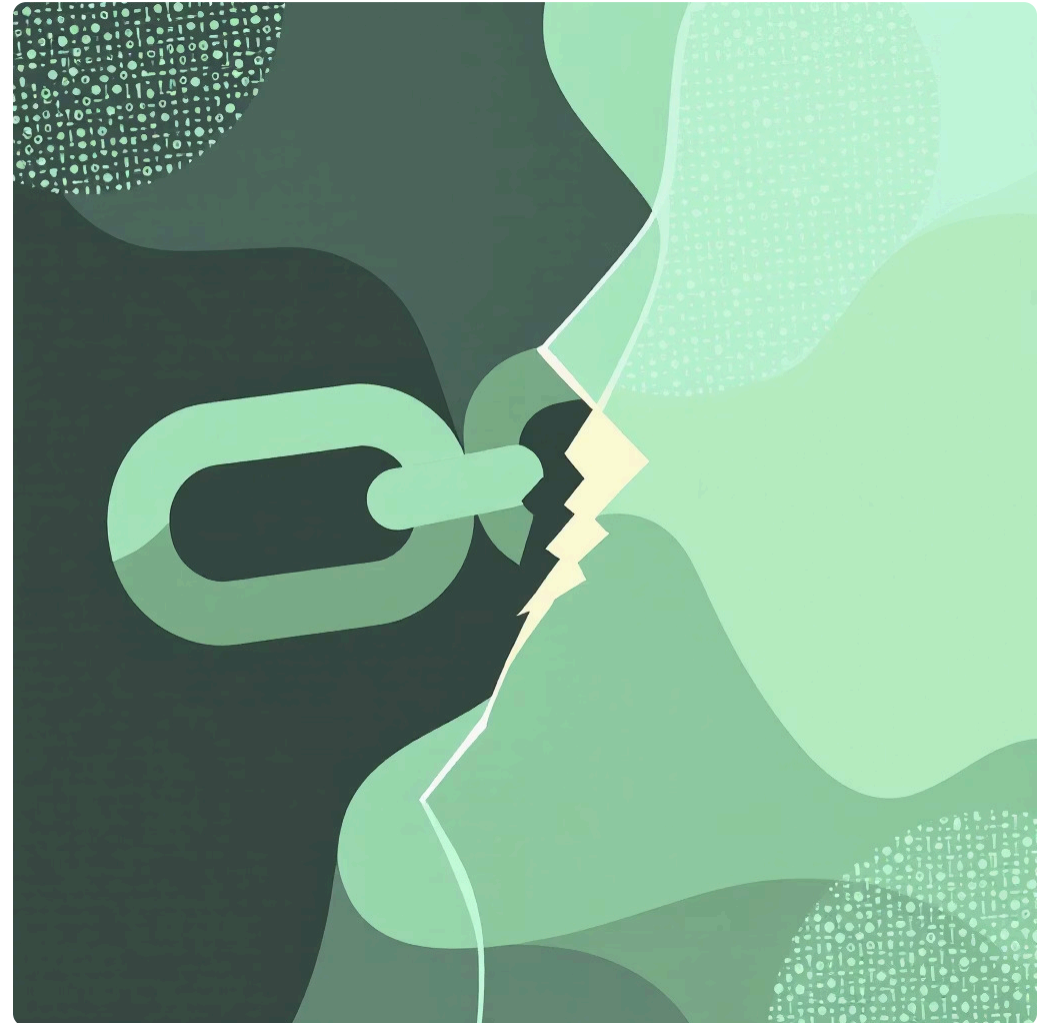
Patient Ownership

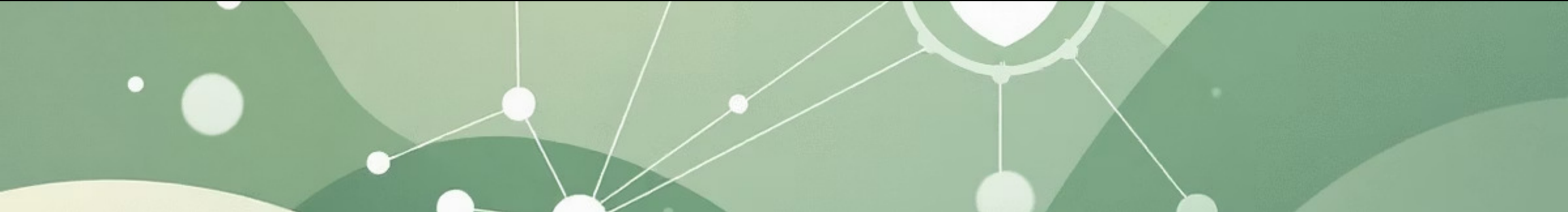
Patients regain full ownership and control over their sensitive health records.

The Challenges of Centralized Healthcare

Traditional healthcare systems, though established, are inherently centralized. This structure introduces significant vulnerabilities and limits patient autonomy.

- High risk of data breaches and manipulation due to single points of failure.
- Patients often lack direct control over their own medical data, leading to trust issues.
- Inefficient data sharing across providers hinders coordinated care.





Project Objectives: Building a Resilient System

VitalX aims to address current limitations by focusing on key strategic objectives.



Secure Record Storage

Implementing robust cryptographic methods for safeguarding medical records.



Patient-Controlled Access

Empowering patients with granular control over who can view their health data.



Immutable Blockchain Logging

Ensuring every transaction and record update is permanently logged on the blockchain.



IPFS for File Storage

Utilizing the InterPlanetary File System (IPFS) for distributed and resilient file storage.

Technology Stack: The Foundation of VitalX

Our robust infrastructure is built upon industry-leading decentralized technologies.



Frontend: React.js

A dynamic and responsive user interface for seamless patient interaction.



Blockchain: Ethereum

The leading smart contract platform, providing security and decentralization.



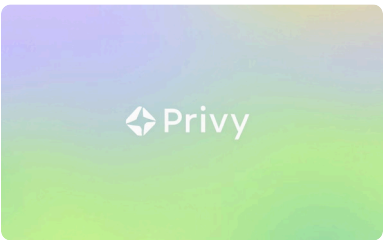
Smart Contracts: Solidity

Programming language for defining the rules and logic on the Ethereum blockchain.



Storage: IPFS (Pinata)

Distributed file storage ensures data availability and censorship resistance, facilitated by Pinata pinning service.

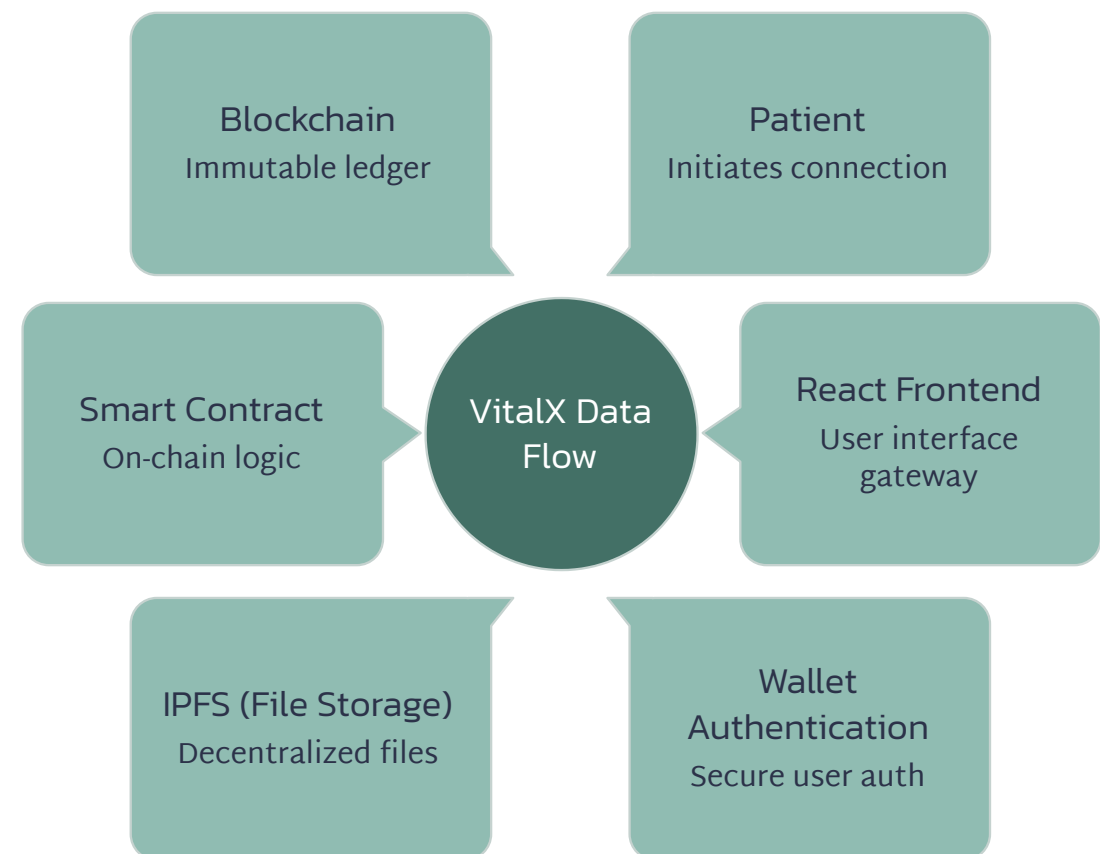
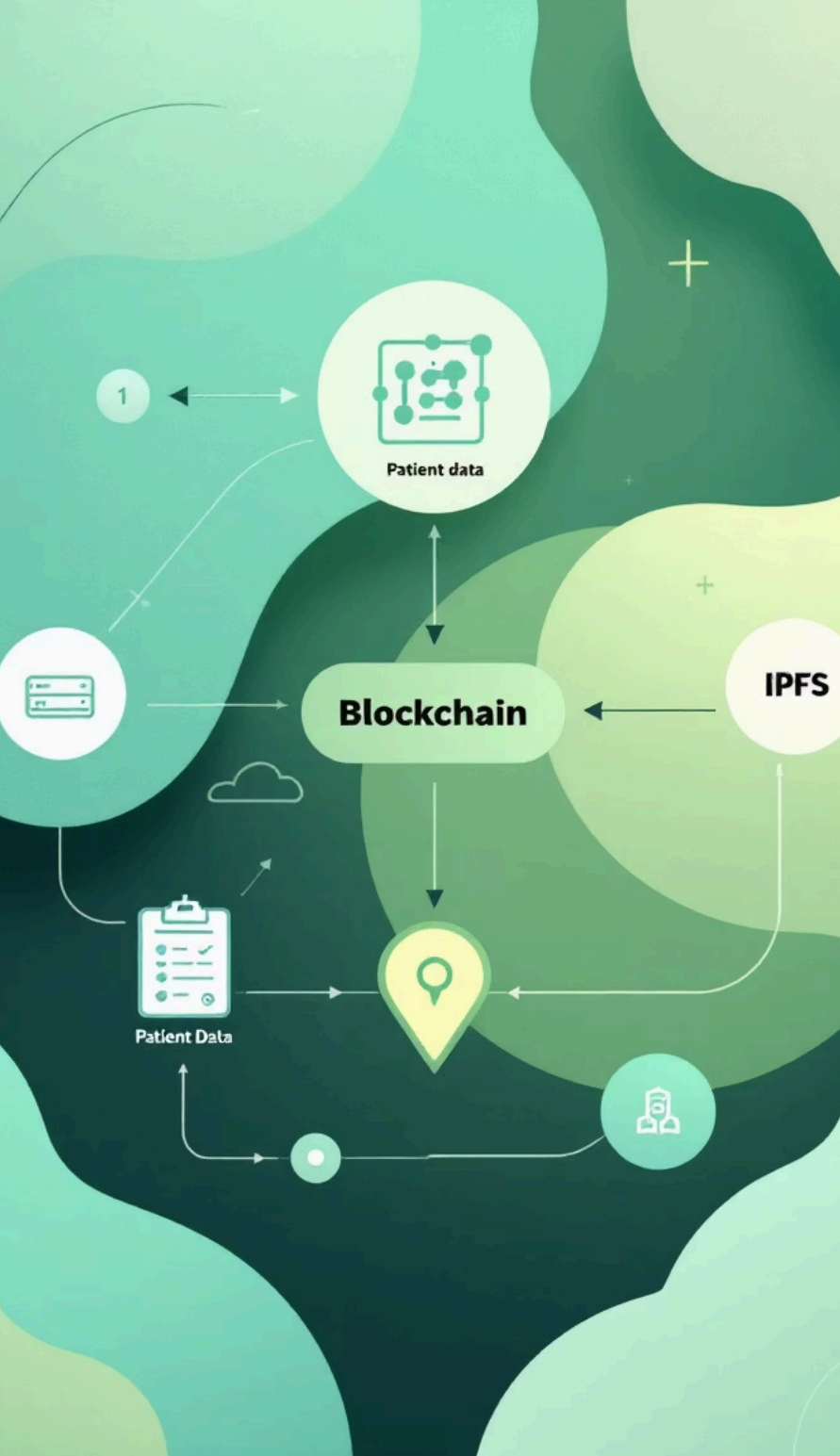


Authentication: Privy Wallet

Secure and user-friendly wallet-based authentication for patient access.

System Architecture: How VitalX Connects

VitalX creates a seamless flow of data, empowering patients through a secure and decentralized ecosystem.



System Modules: Empowering Patients

Patient Module: Your Health, Your Control

1 Wallet Login

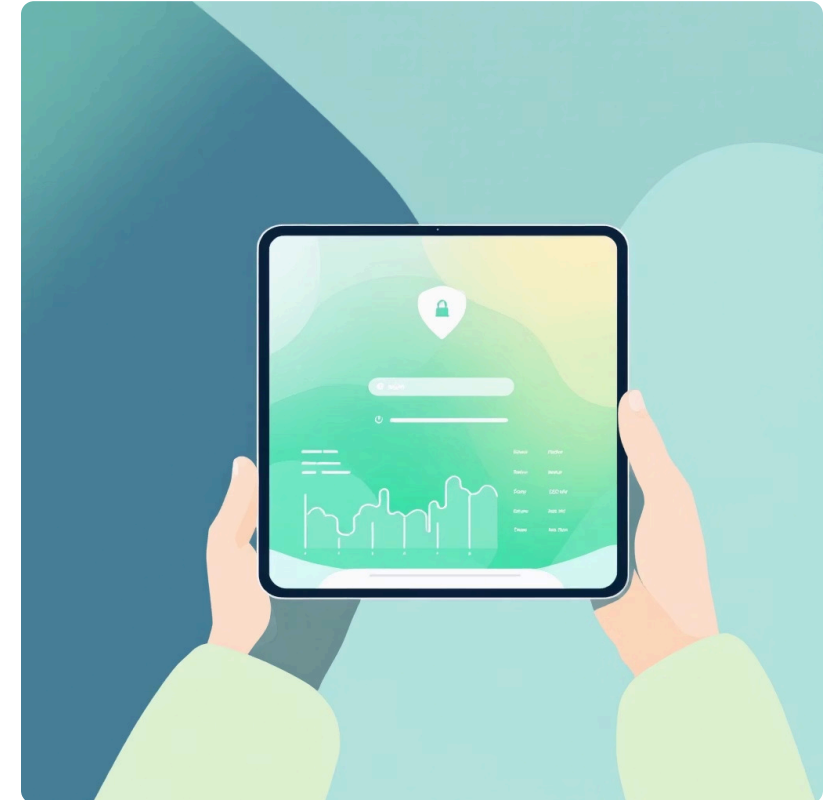
Secure authentication via cryptographic wallets, eliminating traditional usernames and passwords.

2 Upload Medical Records

Patients can securely upload new medical documents, lab results, and diagnostic images.

3 View Stored Records

Easy access to all personal health records, with a clear audit trail of all interactions.



Smart Contract Implementation: The Trust Layer

Our Solidity smart contracts are the backbone of VitalX's security and transparency.



- The smart contract stores only the **IPFS hash** of the medical record and essential metadata, never the sensitive data itself.
- This ensures **immutability and transparency**, as all blockchain entries are permanent and verifiable.
- Records are securely linked to the patient's **unique wallet address**, guaranteeing data ownership and access control.
- Any updates or access events are recorded as new transactions, creating a **complete audit trail**.



VitalX: Advantages & Future Vision

Key Advantages

→ High Security

Leveraging blockchain's cryptographic strength to protect sensitive data.

→ Decentralization

Eliminating single points of failure and increasing system resilience.

→ Data Ownership

Giving patients unprecedented control over their health information.

Future Scope

→ Doctor Access Module

Implementing secure, permissioned access for healthcare providers.

→ Mobile Application

Developing an intuitive mobile app for on-the-go record management.

→ Multi-Chain Support

Expanding compatibility to other blockchain networks for broader adoption.

Conclusion: A New Era for Healthcare Data

VitalX is poised to transform health record management, offering a paradigm shift towards enhanced security and patient empowerment.

Secure & Decentralized

VitalX delivers a robust, decentralized solution for managing healthcare data with blockchain technology.

Improved Privacy

Patients maintain full control and privacy over their sensitive medical information.

Enhanced Trust

The immutable nature of blockchain fosters greater trust and transparency in data handling.